

Cover letter

21 July 2026

Dear Editors of *Classical and Quantum Gravity*,

Please find enclosed the major revision of manuscript CQG-116364, now entitled *Low-lying quasi-normal modes of Kazakov–Solodukhin black holes from Chebyshev and continued-fraction methods*.

The revision addresses every referee comment and substantially strengthens both the physical formulation and the numerical audit. We derive the scalar radial equation explicitly; formulate the odd-parity problem with a gauge-invariant sourced master equation under a clearly stated inverse-Cowling closure; and regenerate all axial results with that effective-source potential. We also give the full candidate-selection algorithm and Schwarzschild seed provenance, continued-fraction-depth and Taylor-order convergence studies, polynomial backward errors, potential and stability audits, and comparison with a pinned external high-precision calculation.

The revised claims are deliberately limited to the evidence. The minimally coupled scalar fundamentals are the principal physical result; first overtones are reported as cross-validated low-lying modes; second overtones are exploratory diagnostics; and the axial frequencies remain conditional on the inverse-Cowling closure. The manuscript does not claim a general high-overtone solver or a unique nonspherical quantum completion of the KS source.

All axial frequencies, figures, tables, shifts, stability diagnostics, and external comparisons in the revised manuscript were regenerated using the inverse-Cowling effective-source potential. No accepted numerical results from the earlier lapse-substitution ansatz remain.

A clean manuscript, a marked line-numbered manuscript, a point-by-point response, the completed revision checklist, and the updated source archive are included. Thank you for considering the revised work.

Yours sincerely,

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